TUTORIAL 13

Gram-Schmidt and Orthogonal Diagonalization

MOTIVATION AND GUIDING IDEA

Gram-Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

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DEFINITION 13.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram-Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j / \|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w \rangle = \langle v, T^* w \rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad RR^T = R^T R = 4I,$$

but $R^2 = -4I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram-Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

MOTIVATION AND GUIDING IDEA

We construct orthonormal bases by systematically subtracting projections and compute projections in polynomial spaces. We study the conditions for orthogonal diagonalization over the reals and unitary diagonalization over the complex numbers, and use an orthonormal eigenbasis to prove normality.

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13.1 Review and Notation

RECALL AND TOOLS

Throughout this tutorial, the inner product space is over $\mathbb{R}$ or $\mathbb{C}$; in the complex case, our convention is:

1. $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0 \iff x = 0$.
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
3. Linearity in the first argument: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.

It follows that the second argument is conjugate linear: $\langle x, ay + bz \rangle = \overline{a}\langle x, y \rangle + \overline{b}\langle x, z \rangle$. This is the same convention as in the preceding tutorial; over $\mathbb{R}$, the two linearity conventions coincide.

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DEFINITION 13.2

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. Vectors $u, v \in V$ are orthogonal if $\langle u, v\rangle = 0$.
2. A list $(e_1, \ldots, e_k)$ is orthonormal if $\langle e_i, e_j\rangle = \delta_{ij}$ for all i, j.
3. For each $x \in V$, define the norm $\|x\| := \sqrt{\langle x, x\rangle}$.

RECALL AND TOOLS

Orthogonal projection onto a line: if $u \neq 0$, the vector

$$\text{proj}_{\text{span}\{u\}}(v) := \frac{\langle v, u\rangle}{\langle u, u\rangle}\, u$$

is the unique vector of the form αu for which $v - \alpha u \perp u$.

13.2 The Gram-Schmidt Algorithm

We present Gram-Schmidt with immediate normalization at each step, constructing an orthonormal list directly.

Let $v_1, \ldots, v_k \in V$ be linearly independent. Define recursively:

1. Set $w_1 := v_1$, and then $e_1 := \dfrac{w_1}{\|w_1\|}$.
2. For $j = 2, \ldots, k$, first define
$$w_j := v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle\, e_i,$$
and then normalize:
$$e_j := \frac{w_j}{\|w_j\|}.$$

The main conclusions are:

1. $(e_1, \ldots, e_k)$ is orthonormal.
2. For every m, $\text{span}\{v_1, \ldots, v_m\} = \text{span}\{e_1, \ldots, e_m\}$.

Formal justification: assume inductively that $e_1, \ldots, e_{j-1}$ are orthonormal. For every $r < j$,

$$\begin{aligned}\langle w_j, e_r\rangle &= \left\langle v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle e_i,\ e_r \right\rangle \\ &= \langle v_j, e_r\rangle - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle \langle e_i, e_r\rangle \\ &= \langle v_j, e_r\rangle - \langle v_j, e_r\rangle = 0.\end{aligned}$$

If $w_j = 0$, then v_j belongs to the span of $e_1, \ldots, e_{j-1}$, which by induction equals the span of $v_1, \ldots, v_{j-1}$. This contradicts the linear independence of the input. Thus $w_j \neq 0$, normalization is permitted, and $\|e_j\| = 1$. Also, $w_j \in \text{span}\{v_1, \ldots, v_j\}$, and the subtraction formula implies $v_j \in \text{span}\{e_1, \ldots, e_{j-1}, w_j\}$. Hence the corresponding prefix spans agree at every step.

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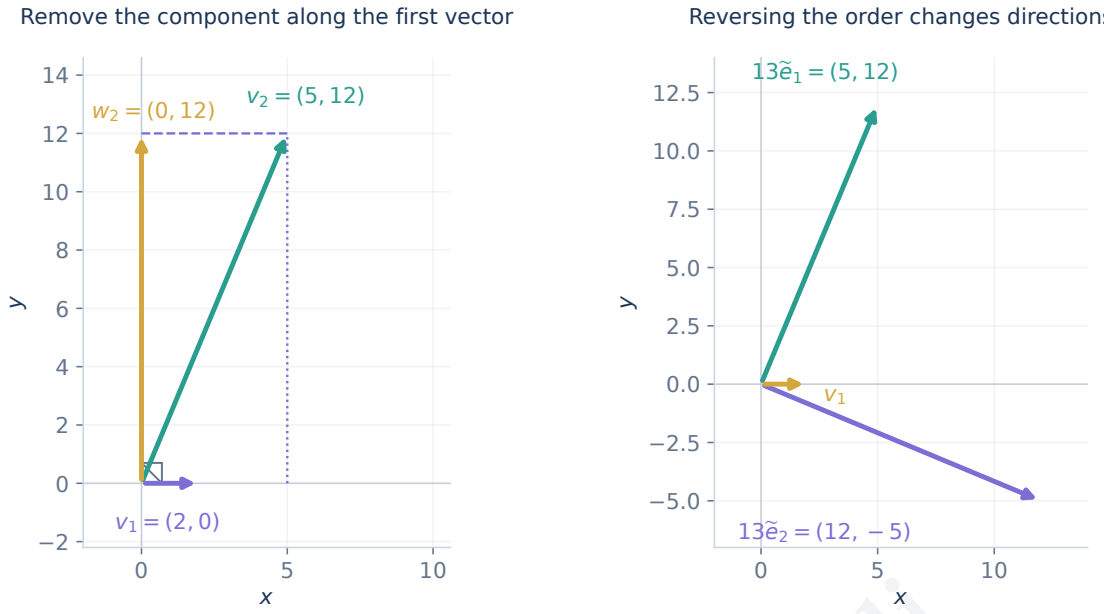


Figure 13.1. At each step, subtract from the next vector its components in the directions already constructed. The residual is orthogonal to the preceding directions; normalization produces the next unit vector.

EXERCISE 13.1

Please apply Gram-Schmidt with normalization to

$$v_1 = (5, 12, 0)^T, \qquad v_2 = (1, 5, 0)^T, \qquad v_3 = (2, -3, 4)^T.$$

Please check that the output is orthonormal.

WORKED SOLUTION

First $\|v_1\| = 13$, so $e_1 = (5, 12, 0)^T/13$. We have

$$\langle v_2, e_1 \rangle = \frac{5 + 60}{13} = 5, \qquad w_2 = v_2 - 5e_1 = \tfrac{1}{13}(-12, 5, 0)^T.$$

The norm of w_2 is 1, so $e_2 = w_2$. At the third step,

$$\langle v_3, e_1 \rangle = \frac{10 - 36}{13} = -2, \qquad \langle v_3, e_2 \rangle = \frac{-24 - 15}{13} = -3.$$

Hence

$$w_3 = v_3 + 2e_1 + 3e_2 = (0, 0, 4)^T, \qquad e_3 = (0, 0, 1)^T.$$

The first two squared norms are $169/169 = 1$, and their inner product is $(-60 + 60)/169 = 0$. Both are orthogonal to e_3, whose norm is also 1.

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EXERCISE 13.2

In $\mathbb{C}^2$ with $\langle x, y\rangle = x^T\overline{y}$, let $v_1 = (4, 3i)^T$ and $v_2 = (2, i)^T$. Please apply Gram-Schmidt with normalization.

WORKED SOLUTION

First $\|v_1\|^2 = 16 + 9 = 25$, so $e_1 = (4, 3i)^T/5$. With linearity in the first slot,

$$\langle v_2, e_1\rangle = 2 \cdot \tfrac{4}{5} + i \cdot \tfrac{-3i}{5} = \tfrac{11}{5}.$$

Thus

$$w_2 = v_2 - \tfrac{11}{5}e_1 = \tfrac{1}{25}(6, -8i)^T, \qquad \|w_2\| = \tfrac{2}{5}, \qquad e_2 = \tfrac{1}{5}(3, -4i)^T.$$

Indeed, $\|e_2\|^2 = (9 + 16)/25 = 1$ and $\langle e_1, e_2\rangle = (12 + (3i)(4i))/25 = 0$.

EXERCISE 13.3

On $\mathbb{R}[t]_{\le 2}$, let $\langle p, q\rangle = \int_{-1}^{1} p(t)q(t)\,dt$. Please apply Gram-Schmidt to $(1, t, t^2)$, normalizing at every step.

WORKED SOLUTION

First $\|1\|^2 = 2$, so $e_1 = 1/\sqrt{2}$. Odd functions integrate to zero on a symmetric interval, hence

$$\langle t, e_1\rangle = 0, \qquad \|t\|^2 = \int_{-1}^{1} t^2 dt = \tfrac{2}{3}, \qquad e_2 = \sqrt{\tfrac{3}{2}}\,t.$$

At the third step,

$$\langle t^2, e_1\rangle = \tfrac{2}{3\sqrt{2}}, \qquad \langle t^2, e_2\rangle = 0, \qquad w_3 = t^2 - \tfrac{1}{3}.$$

Its squared norm is

$$\|w_3\|^2 = \int_{-1}^{1} \left(t^4 - \tfrac{2}{3}t^2 + \tfrac{1}{9}\right) dt = \tfrac{2}{5} - \tfrac{4}{9} + \tfrac{2}{9} = \tfrac{8}{45}.$$

Therefore

$$e_3 = \sqrt{\tfrac{45}{8}}\left(t^2 - \tfrac{1}{3}\right) = \sqrt{\tfrac{5}{8}}(3t^2 - 1).$$

These are the first three Legendre polynomials, normalized for this inner product.

EXERCISE 13.4

In $\mathbb{R}^2$, please compare Gram-Schmidt for the orders (v_1, v_2) and (v_2, v_1), where $v_1 = (2, 0)^T$ and $v_2 = (5, 12)^T$.

WORKED SOLUTION

The first order gives $e_1 = (1, 0)^T$, followed by residual $w_2 = (0, 12)^T$ and $e_2 = (0, 1)^T$. For the reverse order,

$$\widetilde{e}_1 = \tfrac{1}{13}(5, 12)^T, \qquad \langle v_1, \widetilde{e}_1\rangle = \tfrac{10}{13},$$

so

$$\widetilde{w}_2 = (2, 0)^T - \tfrac{10}{169}(5, 12)^T = \tfrac{1}{169}(288, -120)^T.$$

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EXERCISE 13.4 CONTINUED

Its norm is $24/13$, hence $\widetilde{e}_2 = (12, -5)^T/13$. Both outputs are orthonormal and span the same plane, but their directions differ.

EXERCISE 13.5

Let $V = \mathbb{C}[x]_{\leq 3}$ and define

$$\langle f, g \rangle = \sum_{t \in \{-3,-1,1,3\}} f(t)\overline{g(t)}.$$

Let $W = \mathbb{C}[x]_{\leq 2}$. Please prove that this is an inner product, please find an orthonormal basis of W, and please find $W^{\perp}$.

WORKED SOLUTION

Linearity in the first slot and conjugate symmetry follow from the sum. Also $\langle f, f \rangle = \sum_t |f(t)|^2 \geq 0$. Equality gives four distinct roots, so a polynomial of degree at most 3 must vanish identically. Apply Gram-Schmidt to $(1, x, x^2)$. Symmetry gives

$$\sum_t t = \sum_t t^3 = 0, \qquad \sum_t 1 = 4, \qquad \sum_t t^2 = 20, \qquad \sum_t t^4 = 164.$$

Hence

$$e_1 = \tfrac{1}{2}, \qquad e_2 = \tfrac{x}{\sqrt{20}}, \qquad w_3 = x^2 - \frac{20}{4} = x^2 - 5.$$

The values of w_3 are $4, -4, -4, 4$, so $\|w_3\|^2 = 64$ and $e_3 = (x^2 - 5)/8$. Thus (e_1, e_2, e_3) is an orthonormal basis of W. The polynomial

$$u(x) = 5x^3 - 41x$$

is odd, so it is orthogonal to $1, x^2$. Moreover,

$$\langle u, x \rangle = 5 \sum_t t^4 - 41 \sum_t t^2 = 5 \cdot 164 - 41 \cdot 20 = 0.$$

It is nonzero and orthogonal to W. Since $\dim W^{\perp} = 4 - 3 = 1$,

$$W^{\perp} = \operatorname{span}\{5x^3 - 41x\}.$$

Its values are $-12, 36, -36, 12$, so $\|u\|^2 = 2880$; division by $24\sqrt{5}$ normalizes it.

EXERCISE 13.6

Continue with V, W and the preceding inner product. Let P_W be orthogonal projection onto W.

1. Please prove that P_W is linear and $P_W^2 = P_W$.
2. Please compute $P_W(x^3)$ using the orthonormal basis you found.
3. Define $(Tq)(x) = q(x+2) - q(x)$. Please find all $q \in W$ with $Tq \in W^{\perp}$, and please find the unique solution satisfying $\langle q, 1 \rangle = 2$. Please determine whether $\|q\| = 1$ guarantees uniqueness.

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EXERCISE 13.6 CONTINUED

WORKED SOLUTION

1. For the orthogonal decomposition $f = w + u$ with $w \in W$, $u \in W^\perp$, we have $P_W f = w$. Combining the decompositions of f, g gives $P_W(\alpha f + \beta g) = \alpha P_W f + \beta P_W g$. Since P_W is the identity on W, we have $P_W^2 = P_W$.

2. Symmetry gives $\langle x^3, e_1\rangle = \langle x^3, e_3\rangle = 0$, while

$$\langle x^3, e_2\rangle = \frac{164}{\sqrt{20}}.$$

Thus

$$P_W(x^3) = \frac{164}{\sqrt{20}}\frac{x}{\sqrt{20}} = \frac{41}{5}x.$$

The residual $x^3 - P_W(x^3) = (5x^3 - 41x)/5 \in W^\perp$ confirms the answer.

3. For $q(x) = ax^2 + bx + c$,

$$(Tq)(x) = a(x+2)^2 + b(x+2) + c - (ax^2 + bx + c) = 4ax + 4a + 2b.$$

Thus $Tq \in W$. Requiring $Tq \in W^\perp$ forces $Tq = 0$, hence $a = b = 0$. All solutions are constants $q = c$. The condition $\langle q, 1\rangle = 4c = 2$ gives $q = 1/2$. In contrast, $\|q\|^2 = 4|c|^2 = 1$ gives $|c| = 1/2$: there are two real solutions and infinitely many complex ones.

EXERCISE 13.7 EXAM QUESTION

Let $V = \mathbb{R}_4[x]$, the space of real polynomials of degree at most 4, including the zero polynomial. Define $L : V \to \mathbb{R}^3$ by

$$L(p) = (p(0), p(1), p'(1)).$$

Equip V with the inner product

$$\langle p, q\rangle = \int_0^1 p(t)q(t)\,dt + p(0)q(0) + p(1)q(1).$$

Please write $W = \ker L$ and $s(x) = 1 + x + x^2 + x^3 + 2x^4$.

1. Please prove that L is linear.
2. Please compute $\ker L$ and $\operatorname{Im} L$, and please find a basis of each.
3. Please compute the orthogonal projection $P_W(s)$ for the given inner product.

WORKED SOLUTION

1. Let $p, q \in V$ and $a, b \in \mathbb{R}$. By the linearity of evaluation and differentiation,

$$\begin{aligned} L(ap + bq) &= ((ap + bq)(0), (ap + bq)(1), (ap + bq)'(1)) \\ &= (ap(0) + bq(0), ap(1) + bq(1), ap'(1) + bq'(1)) \\ &= aL(p) + bL(q). \end{aligned}$$

so L is linear.

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EXERCISE 13.7 EXAM QUESTION CONTINUED

2. The condition $p \in \ker L$ is equivalent to the three conditions

$$p(0) = 0, \qquad p(1) = 0, \qquad p'(1) = 0.$$

The first says that x divides p. The last two say that 1 is a root of multiplicity at least 2: if $p(x) = (x-1)q(x)$, then $p'(x) = q(x) + (x-1)q'(x)$, so $p'(1) = q(1)$. Thus $(x-1)^2$ divides p. Since x and $(x-1)^2$ have no common root, and p has degree at most 4,

$$W = \{x(x-1)^2(a+bx) : a, b \in \mathbb{R}\}.$$

Define

$$w_1 = x(x-1)^2 = x^3 - 2x^2 + x, \qquad w_2 = x^2(x-1)^2 = x^4 - 2x^3 + x^2.$$

The polynomials w_1, w_2 span W and are linearly independent, for example because they have different degrees. Hence (w_1, w_2) is a basis of W, and $\dim W = 2$.

To determine the image, exhibit polynomials mapped to the three standard basis vectors:

$$h_1 = (x-1)^2, \qquad h_2 = 2x - x^2, \qquad h_3 = x^2 - x.$$

A direct calculation gives

$$L(h_1) = (1,0,0), \qquad L(h_2) = (0,1,0), \qquad L(h_3) = (0,0,1).$$

Thus $\operatorname{Im} L = \mathbb{R}^3$, with the standard basis as a basis of the image. As a dimension check, $\dim \ker L + \dim \operatorname{Im} L = 2 + 3 = 5 = \dim V$.

3. Seek $p = P_W(s)$ in the form $p = aw_1 + bw_2$. The definition of orthogonal projection requires $p \in W$ and $s - p \in W^\perp$. Since w_1, w_2 form a basis of W, it suffices to require

$$\langle s - p, w_1 \rangle = 0, \qquad \langle s - p, w_2 \rangle = 0.$$

Every $w \in W$ satisfies $w(0) = w(1) = 0$, so the endpoint terms in the inner product vanish whenever one factor is w_1 or w_2. First compute the inner products between the basis vectors:

$$\begin{aligned}
\langle w_1, w_1 \rangle &= \int_0^1 (t^2 - 4t^3 + 6t^4 - 4t^5 + t^6)\, dt \\
&= \frac{1}{3} - 1 + \frac{6}{5} - \frac{2}{3} + \frac{1}{7} = \frac{1}{105}, \\
\langle w_1, w_2 \rangle &= \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7)\, dt \\
&= \frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{1}{280}, \\
\langle w_2, w_2 \rangle &= \int_0^1 (t^4 - 4t^5 + 6t^6 - 4t^7 + t^8)\, dt \\
&= \frac{1}{5} - \frac{2}{3} + \frac{6}{7} - \frac{1}{2} + \frac{1}{9} = \frac{1}{630}.
\end{aligned}$$

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EXERCISE 13.7 EXAM QUESTION CONTINUED

Multiplying out the polynomials also gives

$$\langle s, w_1\rangle = \int_0^1 (2t^7 - 3t^6 + t^5 - t^2 + t)\,dt = \frac{2}{8} - \frac{3}{7} + \frac{1}{6} - \frac{1}{3} + \frac{1}{2} = \frac{13}{84},$$
$$\langle s, w_2\rangle = \int_0^1 (2t^8 - 3t^7 + t^6 - t^3 + t^2)\,dt = \frac{2}{9} - \frac{3}{8} + \frac{1}{7} - \frac{1}{4} + \frac{1}{3} = \frac{37}{504}.$$

The orthogonality conditions are therefore

$$\begin{pmatrix} 1/105 & 1/280 \\ 1/280 & 1/630 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 13/84 \\ 37/504 \end{pmatrix}.$$

Multiplication by 2520 gives the system

$$24a + 9b = 390, \qquad 9a + 4b = 185.$$

Multiply the first equation by 4 and the second by 9, and subtract:

$$(96a + 36b) - (81a + 36b) = 1560 - 1665 \quad\Longrightarrow\quad 15a = -105.$$

Hence $a = -7$; the second equation gives $-63 + 4b = 185$, so $b = 62$. We conclude that

$$\begin{aligned} P_W(s) &= -7x(x-1)^2 + 62x^2(x-1)^2 \\ &= 62x^4 - 131x^3 + 76x^2 - 7x. \end{aligned}$$

The polynomial belongs to W by the factorization in the first line, and the two inner product equations ensure that its difference from s is orthogonal to all of W. These are exactly the two conditions that uniquely characterize the orthogonal projection.

13.3 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION 13.3

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w\rangle = \langle v, T^*w\rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^{\,T}$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

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PROPOSITION 13.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram-Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

EXERCISE 13.8

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Please prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \ldots, e_n)$ and real numbers $h_1, \ldots, h_n$ such that

$$He_j = h_je_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_je_j.$$

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EXERCISE 13.8 CONTINUED

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \operatorname{diag}(h_1, \ldots, h_n), \qquad [K]_B = D_K := \operatorname{diag}(k_1, \ldots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H+K}{2}, \quad T^* = \frac{H-K}{2}.$$

Compute the two products separately, without assuming in advance that the factors commute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H+K)(H-K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.

We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \ldots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

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EXERCISE 13.8 CONTINUED

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}} d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$AA^* = (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*,$$
$$A^*A = (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*.$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

Additional practice

EXERCISE 13.9 FOUNDATION

In $\mathbb{R}^3$ with the standard inner product, consider the list

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (3, 3, 0)^T, \qquad v_3 = (1, 0, 1)^T.$$

Please apply Gram-Schmidt, explicitly handling any zero residual. Please find and please check an orthonormal basis of the input span.

WORKED SOLUTION

First, $\|v_1\| = \sqrt{2}$, so

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T.$$

For the second input,

$$\langle v_2, e_1 \rangle = 3\sqrt{2}, \qquad w_2 = v_2 - 3\sqrt{2}\, e_1 = 0.$$

We cannot normalize w_2; we skip it because $v_2 = 3v_1$. For the third input,

$$\langle v_3, e_1 \rangle = \frac{1}{\sqrt{2}}, \qquad w_3 = v_3 - \frac{1}{\sqrt{2}} e_1 = (1/2, -1/2, 1)^T.$$

We have $\|w_3\|^2 = 1/4 + 1/4 + 1 = 3/2$, so the second output vector is

$$e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

A direct check gives

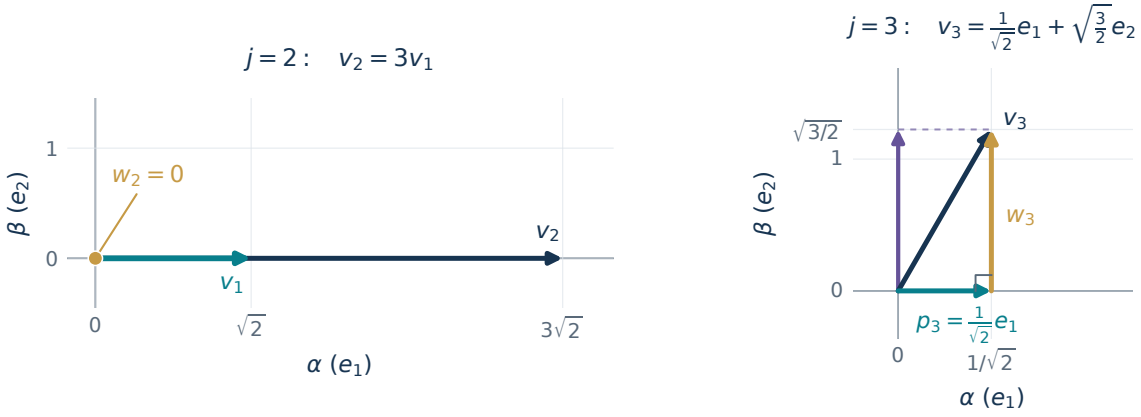
$$\|e_1\|^2 = \frac{1+1}{2} = 1, \qquad \|e_2\|^2 = \frac{1+1+4}{6} = 1, \qquad \langle e_1, e_2 \rangle = \frac{1-1}{\sqrt{12}} = 0.$$

The vectors v_1, v_3 are linearly independent: in $av_1 + bv_3 = 0$, the third coordinate gives $b = 0$, and then $a = 0$. Thus the input span has dimension 2, and (e_1, e_2) is its orthonormal basis. It cannot contain three orthonormal vectors, since those vectors would be linearly independent.

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$$x = \alpha e_1 + \beta e_2, \qquad e_1 = \frac{(1,1,0)^T}{\sqrt{2}}, \quad e_2 = \frac{(1,-1,2)^T}{\sqrt{6}}$$



$$v_2 - \langle v_2, e_1 \rangle e_1 = 0 \qquad v_3 - p_3 = w_3 = \sqrt{\tfrac{3}{2}}\, e_2$$

Figure 13.2. The axes α, β are coordinates in the orthonormal basis (e_1, e_2) of the plane $\mathrm{span}\{v_1, v_3\}$: each point represents $x = \alpha e_1 + \beta e_2$. The input coordinates in this chart are $(\sqrt{2}, 0)^T$, $(3\sqrt{2}, 0)^T$, and $(1/\sqrt{2}, \sqrt{3/2})^T$, respectively. On the left, $v_2 = 3v_1$, so the residual $w_2 = 0$ must be skipped. On the right, subtract $p_3 = \langle v_3, e_1 \rangle e_1$ from v_3; the residual $w_3 = (1/2, -1/2, 1)^T = \sqrt{3/2}\, e_2$ is orthogonal to e_1.

EXERCISE 13.10 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Please prove that there is an orthogonal matrix Q for which both $Q^T AQ$ and $Q^T BQ$ are diagonal if and only if $AB = BA$.

2. Please find such a matrix for

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix},$$

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^T AQ = D_A$ and $Q^T BQ = D_B$ are diagonal, then $A = QD_A Q^T$ and $B = QD_B Q^T$. Since diagonal matrices commute,

$$AB = QD_A D_B Q^T = QD_B D_A Q^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y \rangle = \langle x, By \rangle,$$

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EXERCISE 13.10 CHALLENGE CONTINUED

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = -q_1, \quad Aq_2 = -q_2, \quad Aq_3 = 4q_3, \qquad Bq_1 = 7q_1, \quad Bq_2 = 3q_2, \quad Bq_3 = 6q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^T AQ = \operatorname{diag}(-1,-1,4), \qquad Q^T BQ = \operatorname{diag}(7,3,6),$$
$$Q^T ABQ = \operatorname{diag}(-7,-3,24).$$

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EXERCISE 13.11 CHALLENGE

Let $v_1, \dots, v_k$ be independent in a real inner product space, and define

$$u_i = \sum_{j=1}^{i} a_{ji} v_j, \qquad a_{ii} > 0.$$

Please prove that normalized Gram-Schmidt produces the same orthonormal list from $(u_1, \dots, u_k)$ and $(v_1, \dots, v_k)$. Please then allow any $a_{ii} \neq 0$ and please describe the relation between the two outputs.

WORKED SOLUTION

For each i, the first i vectors of the two lists span the same subspace. One inclusion follows from the definition; the reverse follows by successively solving for v_i using $a_{ii} \neq 0$. In particular the u_i are independent. Let e_i be the output for the v_i, with residual

$$r_i = v_i - \sum_{j<i} \langle v_i, e_j \rangle e_j, \qquad e_i = r_i / \|r_i\|.$$

Both Gram-Schmidt runs subtract the orthogonal projection onto the same previous span. The terms $a_{ji} v_j$ for $j < i$ lie in that span and disappear from the residual of u_i. Its residual is therefore $a_{ii} r_i$. If $a_{ii} > 0$, normalization gives exactly e_i. For arbitrary nonzero diagonal coefficients, normalization instead gives

$$f_i = \frac{a_{ii} r_i}{|a_{ii}| \|r_i\|} = \frac{a_{ii}}{|a_{ii}|} e_i.$$

Earlier sign changes do not affect orthogonal projections or previous spans, so this relation holds at every step.

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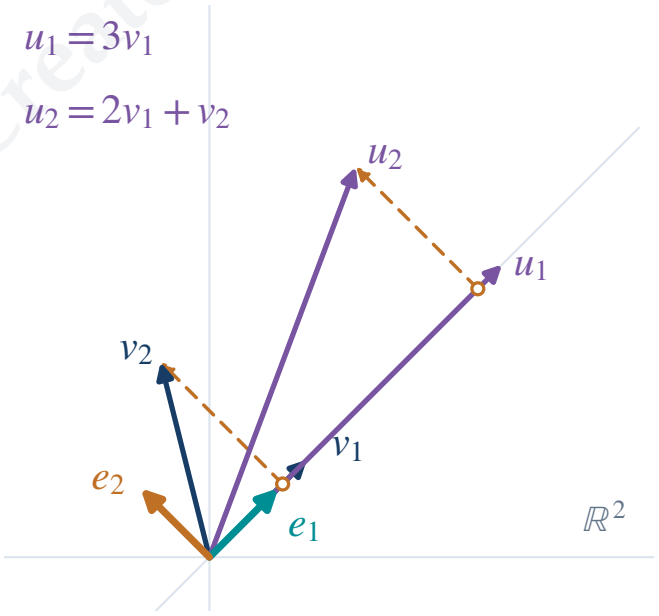


Figure 13.3. Here $u_1 = 3v_1$ and $u_2 = 2v_1 + v_2$. Both lists give the same unit vectors e_1, e_2 under Gram--Schmidt. Dashed segments show the second-step residuals.

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EXERCISE 13.12 CHALLENGE

Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite, meaning $x^T Ax > 0$ for every $x \neq 0$. Please prove that there is a unique symmetric positive definite matrix B satisfying $B^2 = A$.

WORKED SOLUTION

The spectral theorem gives an orthogonal matrix Q with

$$A = QDQ^T, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Every λ_j is positive, because a corresponding unit eigenvector q_j satisfies $\lambda_j = q_j^T Aq_j > 0$. Define

$$B = Q \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})Q^T.$$

It is symmetric and satisfies $B^2 = A$. For $x \neq 0$, put $y = Q^T x \neq 0$. Then $x^T Bx = \sum_j \sqrt{\lambda_j} y_j^2 > 0$. For uniqueness, let C be another symmetric positive definite square root. Since $CA = C^3 = AC$, it preserves every eigenspace $E_\lambda(A)$. Its restriction to that eigenspace is symmetric and positive definite, so it has an orthonormal eigenbasis. For a vector with $Cv = \mu v$, positivity gives $\mu > 0$, and

$$\mu^2 v = C^2 v = Av = \lambda v.$$

Thus $\mu = \sqrt{\lambda}$. On each eigenspace of A, both B and C act as multiplication by $\sqrt{\lambda}$. These eigenspaces span $\mathbb{R}^n$, proving $C = B$.

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